



Complete Guide

TypeScript - Intro & install

Programming Language:

- * Programming language is a set of rules, keywords, and syntax used to write instructions that a computer can understand and execute to perform specific tasks.
- * Humans use English, Tamil, or Hindi to communicate with each other, programmers use-programming languages such as Java, Python, TypeScript, or C# to communicate with computers.

TypeScript

- * Author of Typescript ANDERS HELJSBERG, The lead architect of C#.
- * Public announcement Oct 2012 developed by MICROSOFT.

What is TypeScript?

- * File Name ends with (.ts).
- * It is a syntactic superset of JavaScript, which adds static typing, Means it adds TYPE syntax on top of JavaScript.
- * allows developers to add TYPES.
- * TypeScript only exist in compile time.
- * Typescript uses all features in the JavaScript.

How TypeScript works?

- * Write Codes in Typescript, Extension (.ts)
- * The Typescript compiler converts it into JavaScript.
- * The Codes will run in any JavaScript Environment (browser, node.js).

Typescript -> TypeScript Compiler -> JavaScript

JS vs TS

- * JS codes will work in TS too.
- * TS adds optional feature (i.e. TYPE) to make coding Easier and Safer.
- * Helps catch Errors in codes before Execution.
- * Large project managed Easier.

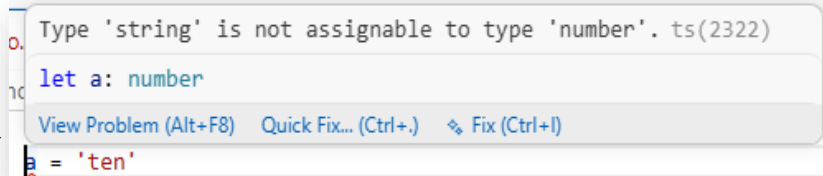
Warning before execution:

i.e.

```
var a = 5;  
var a = 'five';
```

JavaScript → No Error,

Typescript →



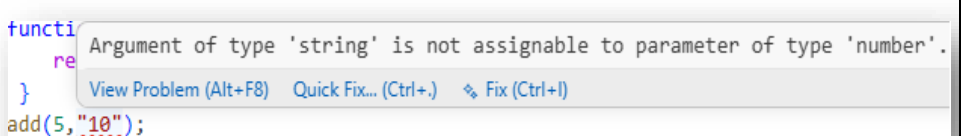
i.e.

```
function add (a,b) {  
    return a+b;  
}  
add(5,"10");
```

Output: 510 instead of 15 (Type Error)

JavaScript → No Error

Typescript →



Installation Typescript

NPM - Node Package Manager

NPX - Node Package Executer

TSC - TypeScript Compiler

TSX - TypeScript Executer

Requirement:

- * Node.js
- * TypeScript Compiler
 - * npm install -g typescript
- * TypeScript Executer
 - * npm install -g tsx
- * VSCode editor
- * create a folder for files.

Execute in Command Prompt/Terminal (run code):

open command prompt in the folder which is used save your TypeScript files.

- * direct execution
 - * tsx (fileName.ts)
 - * tsx (SubFoldName)/(fileName.ts)
- * create JavaScript file of the TyeScript file then executes the JavaScript file.
 - If no subfolder:
 - * tsc (FileName.ts) ➕ node (FileName.js)
 - If the file is in subfolder:
 - * tsc (SubFoldName)/(FileName.ts) ➕ node (SubFoldName)/(FileName.js)
- * Error occurd when you run tsc command for the first time (cannot be loaded error)
 - * Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned

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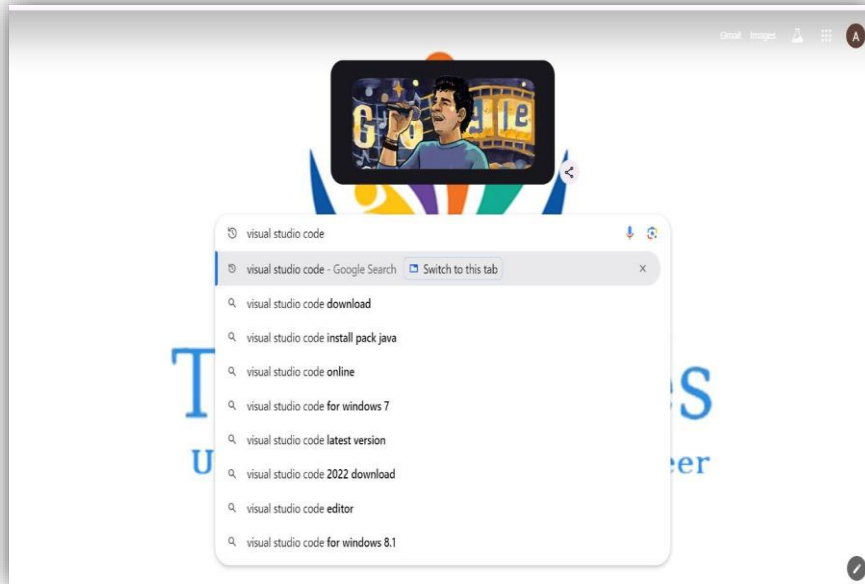
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*Installation Process Starts  
From Next Page*

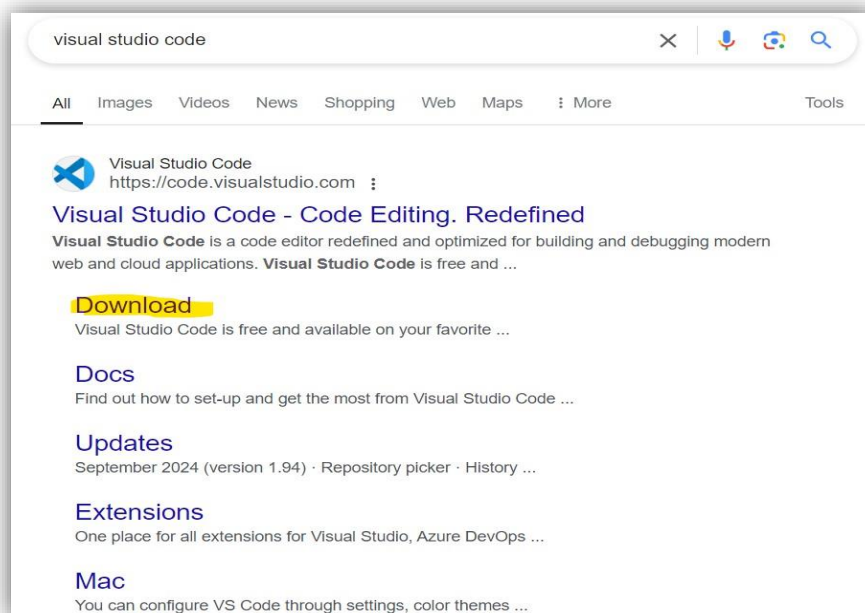
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## VISUAL STUDIO CODE DOWNLOAD & INSTALLATION

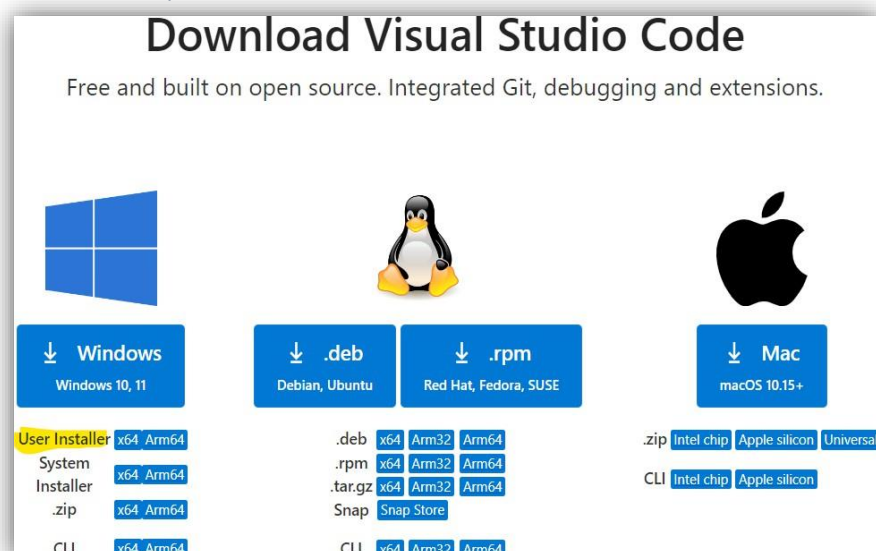
STEP 1: Open Google Chrome and type visual studio code and click enter.



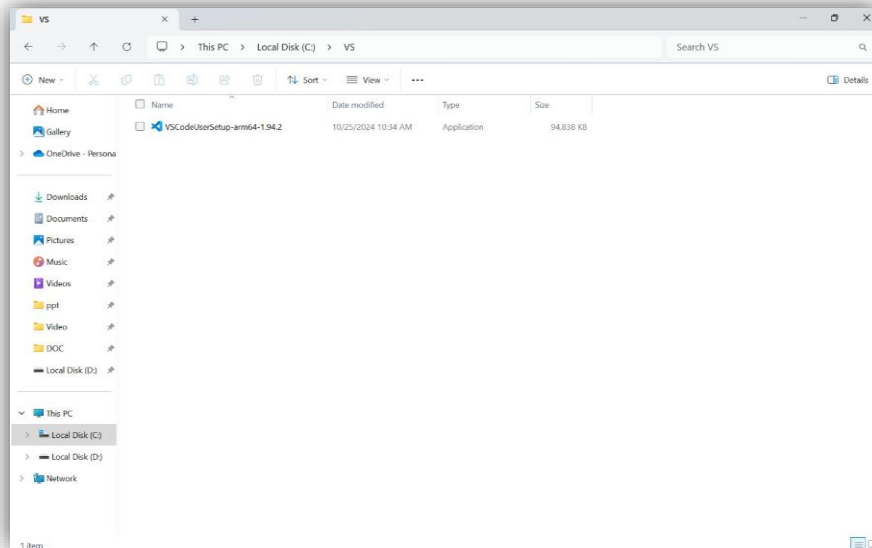
STEP 2: Click on the Download.



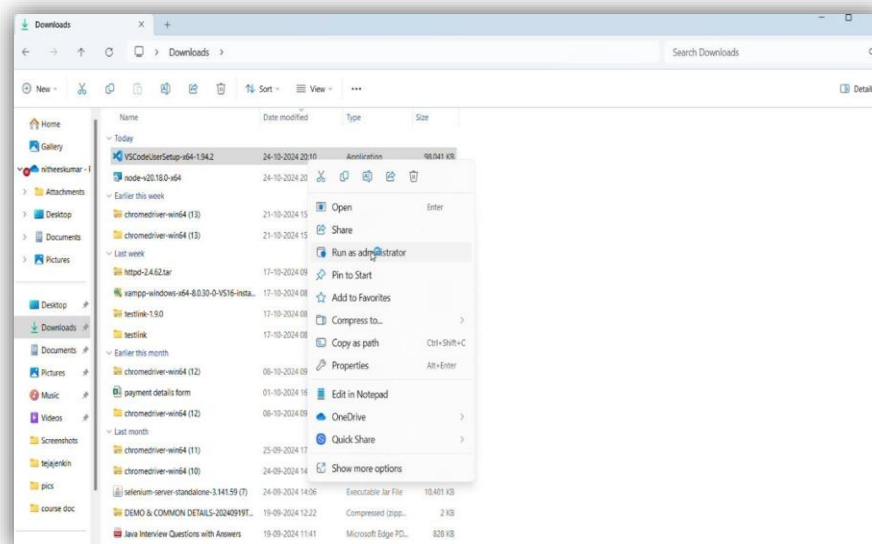
STEP 3: Choose Based on your machines.



STEP 4: It takes you to the folder where visual studio code is saved, right click on it and copy the files and store it in C Drive

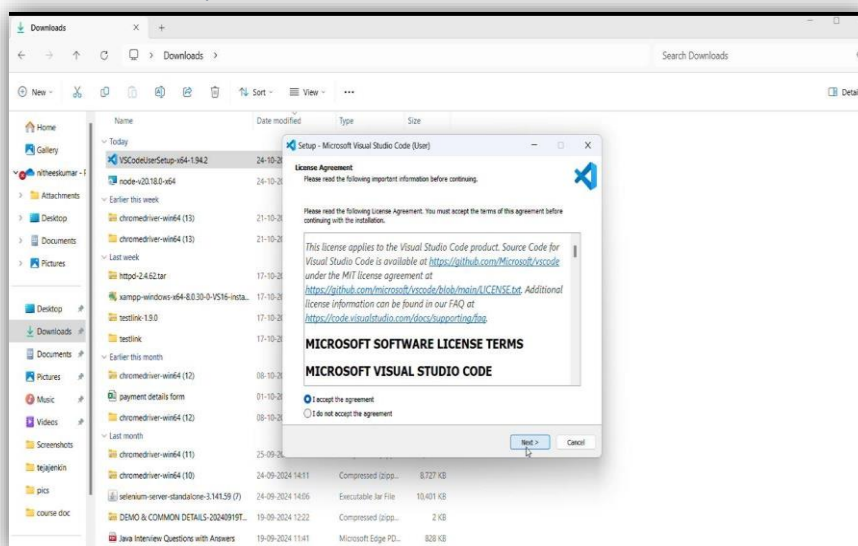


STEP 5: Then right click on it, and choose run as administrator option

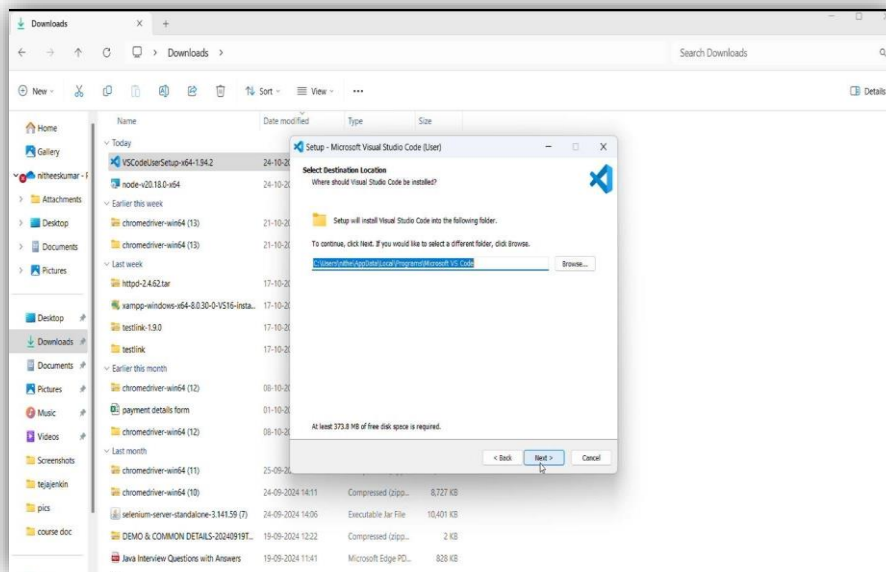


**Note:** It will ask you whether you want the app to allow changes click on yes

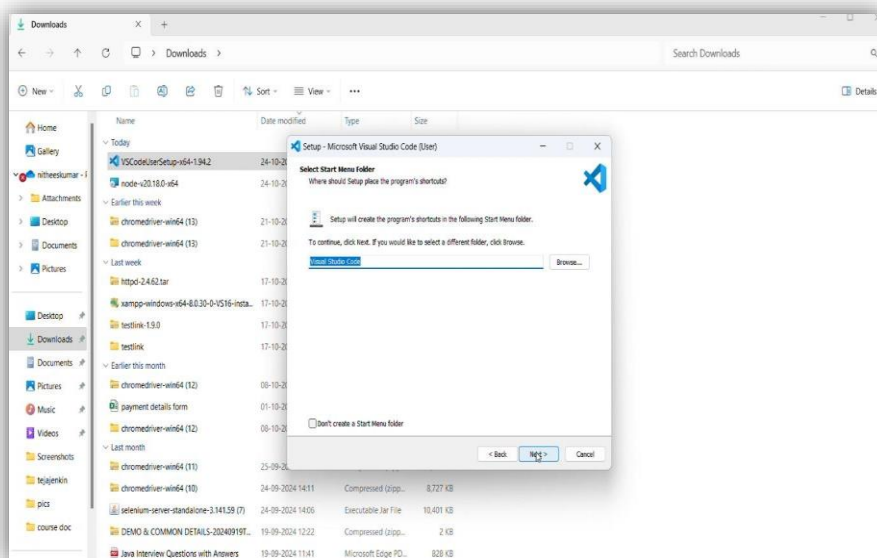
STEP6: After which it will ask you for installation process, where select 'I accept the agreement' and click next option



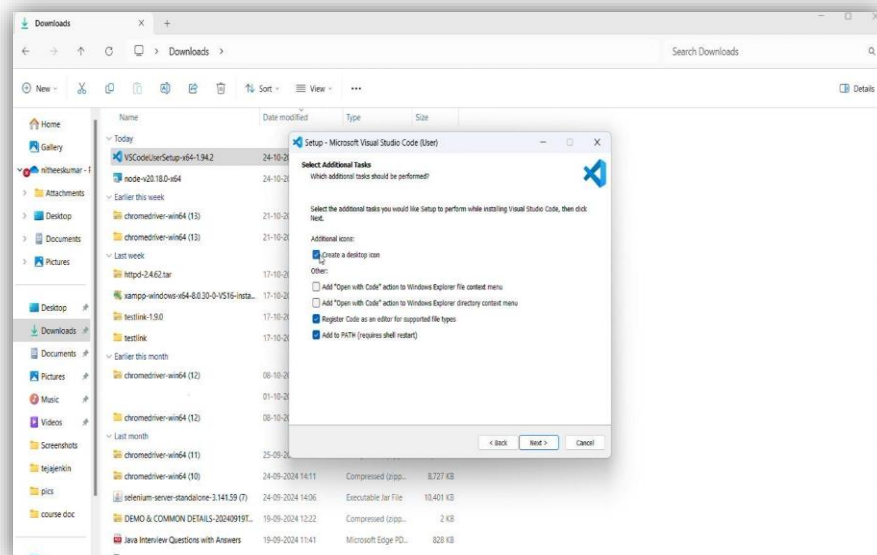
STEP7: Click on the next option.



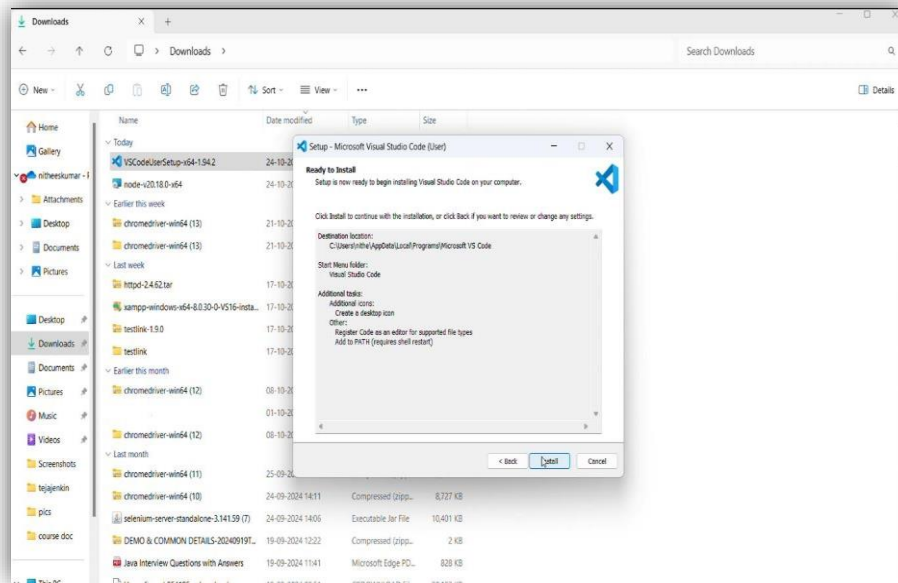
STEP 8: Click on the next option.



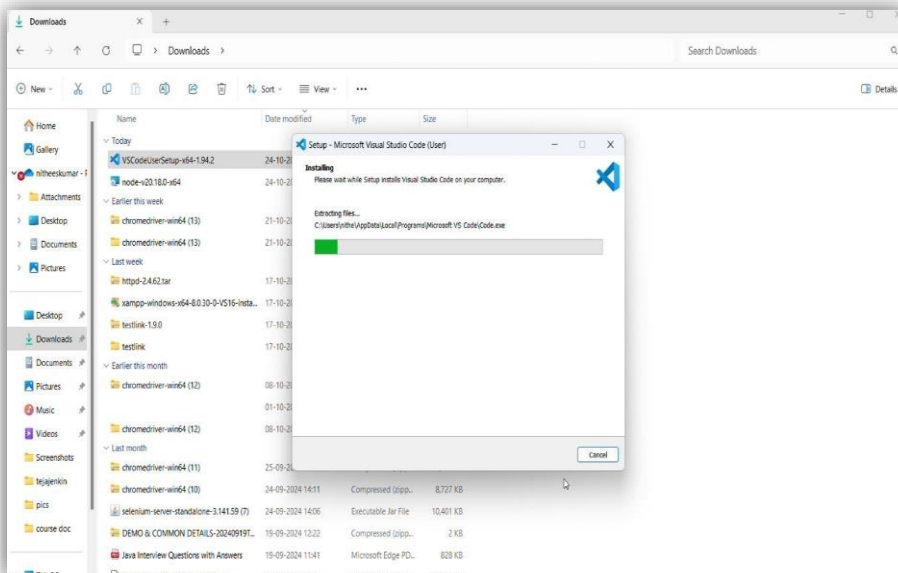
STEP 9: Click on the Create a Desktop Icon and select next option.



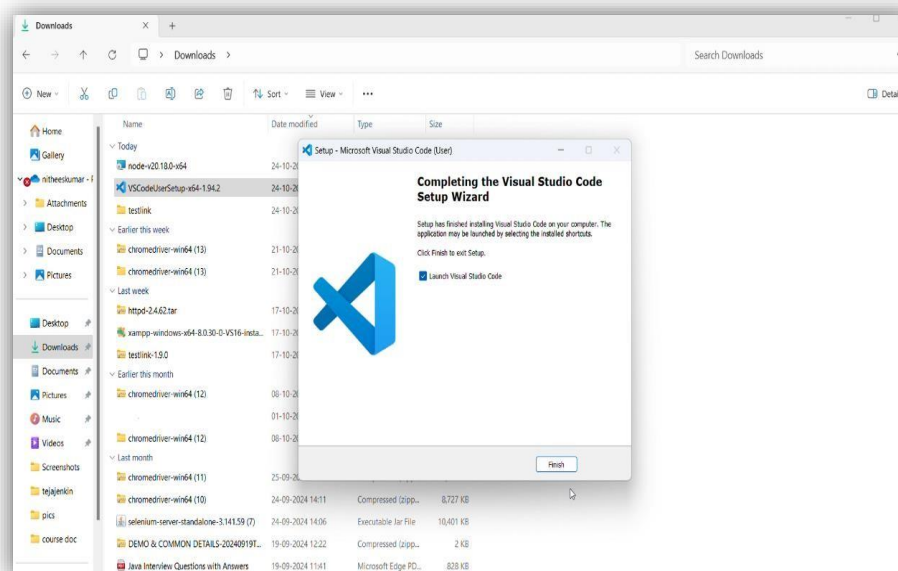
STEP 10: Click on install button



STEP 11: Visual studio installing is in progress.



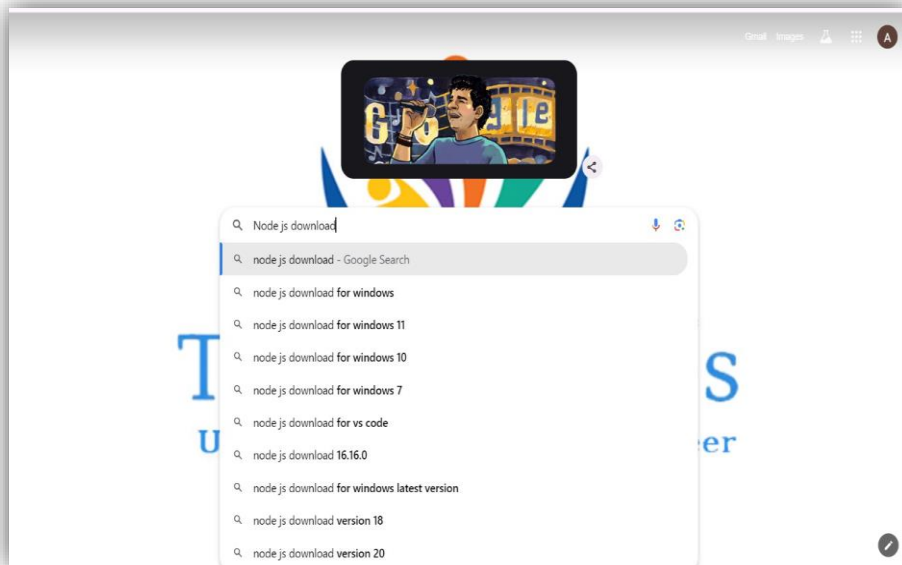
STEP 12: Click Finish to complete the Installation & Create a shortcut to the desktop



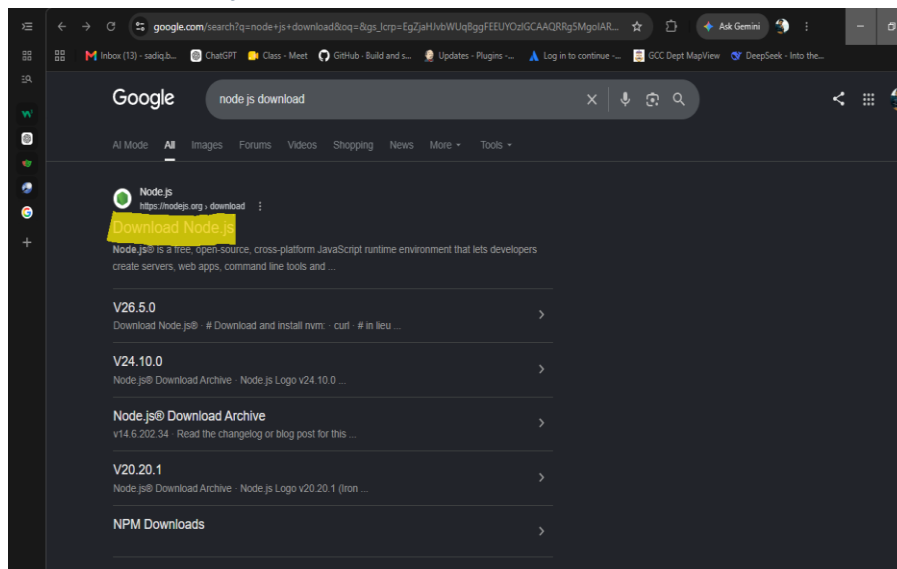


## NODE JS DOWNLOAD & INSTALLATION

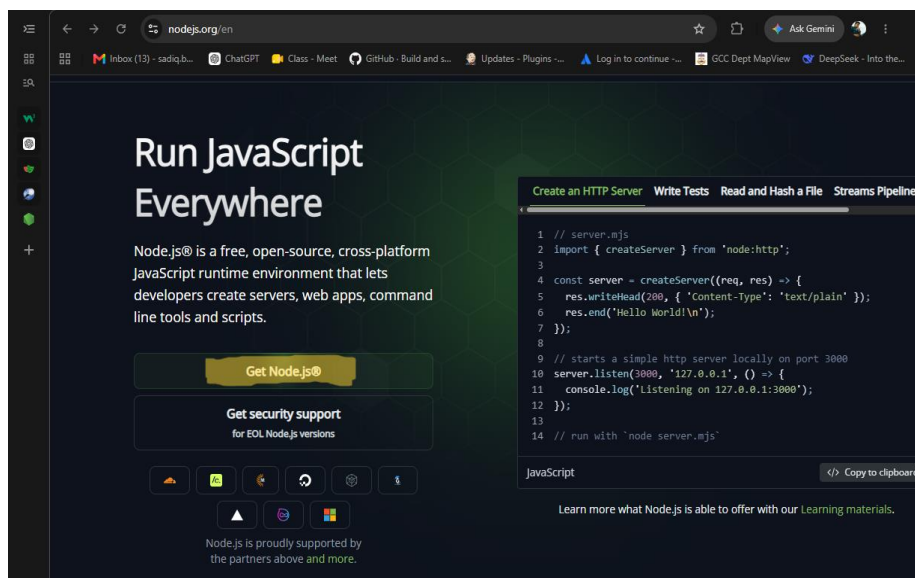
STEP 1: Open google chrome and type Node Js download



STEP 2: Click on Download Node.js.

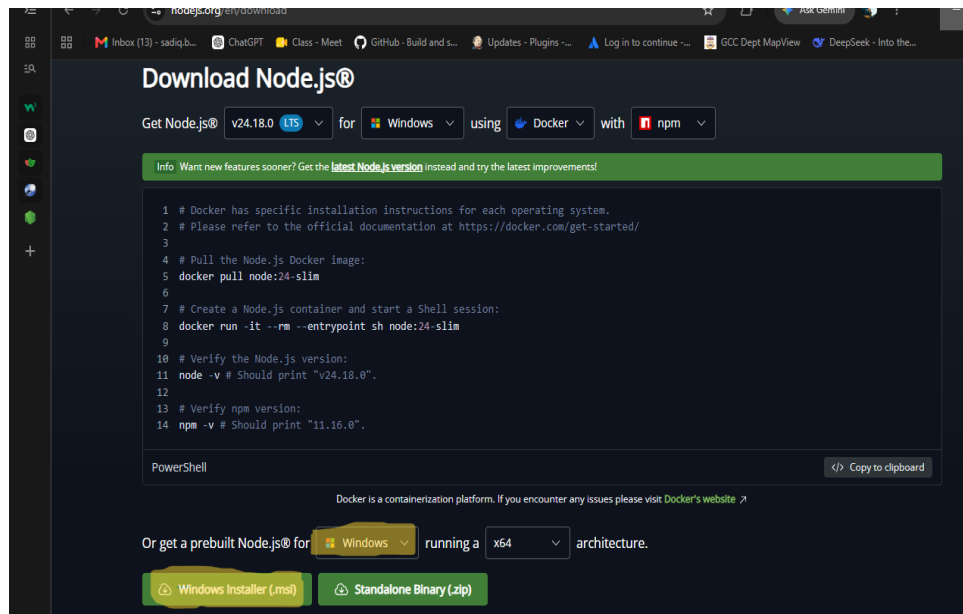


STEP 3: Click on Download option, where Node file was downloaded.

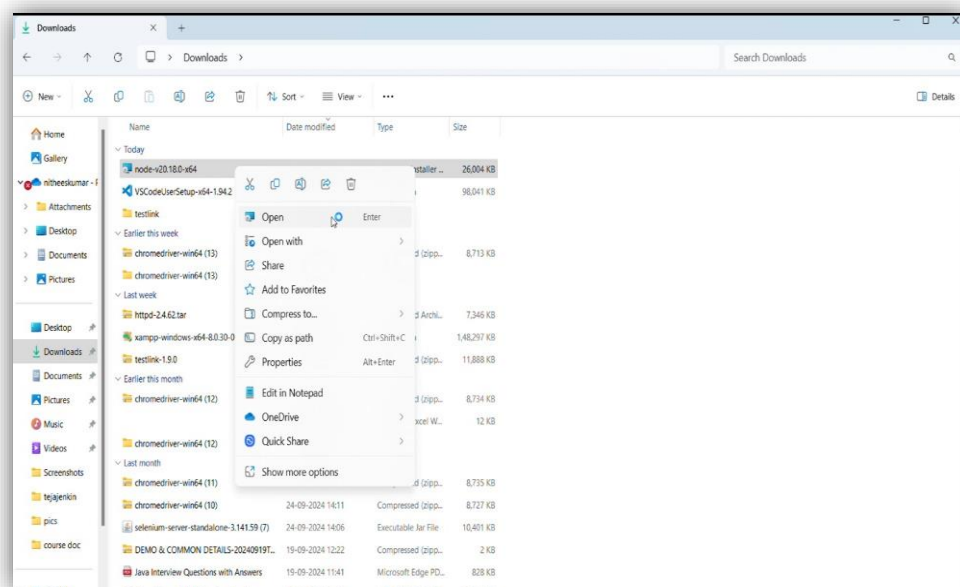




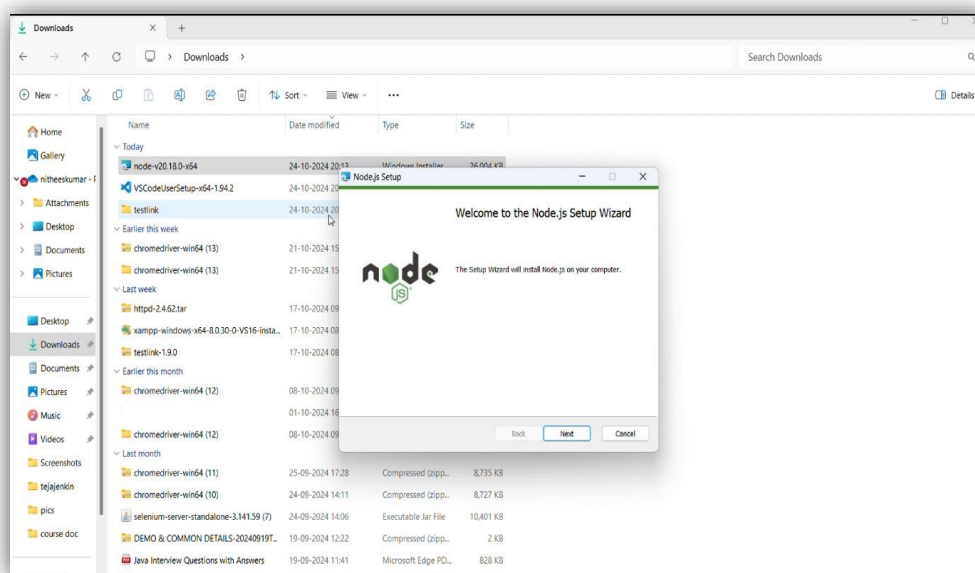
## STEP 4: Select OS type of the System and Click Windows Installer (.msi) file



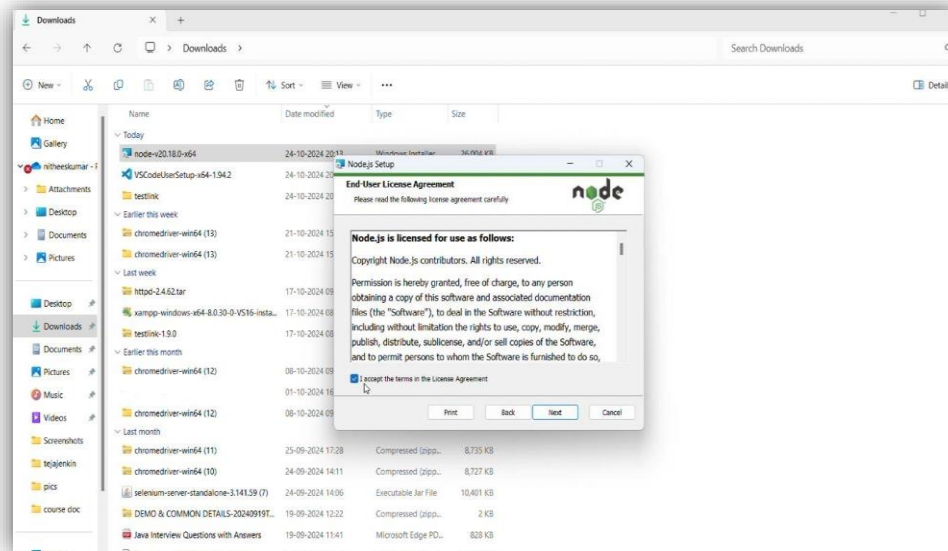
## STEP 5: Go to the Downloaded file And Right Click Open the Node.js file



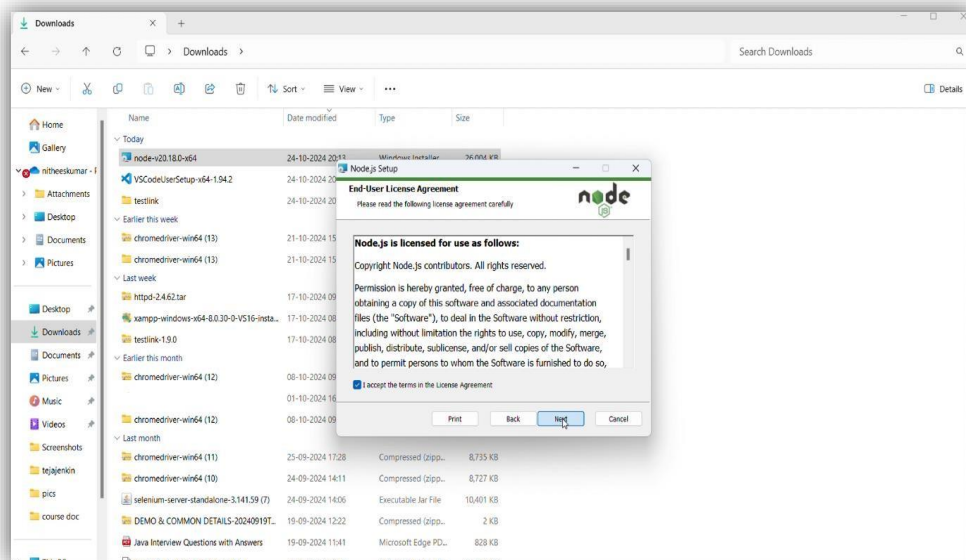
## STEP 6: Click next to continue.



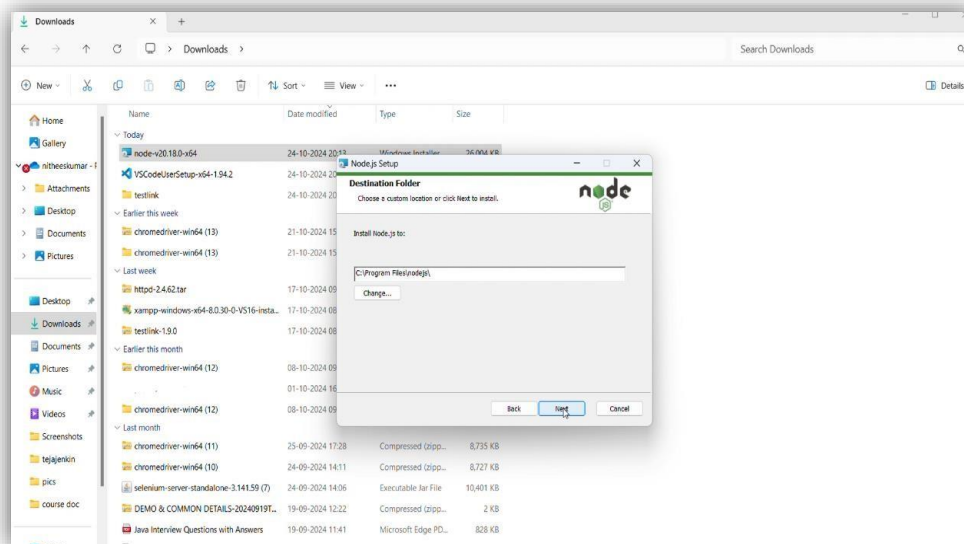
STEP 7: Click on the checkbox and select next option.



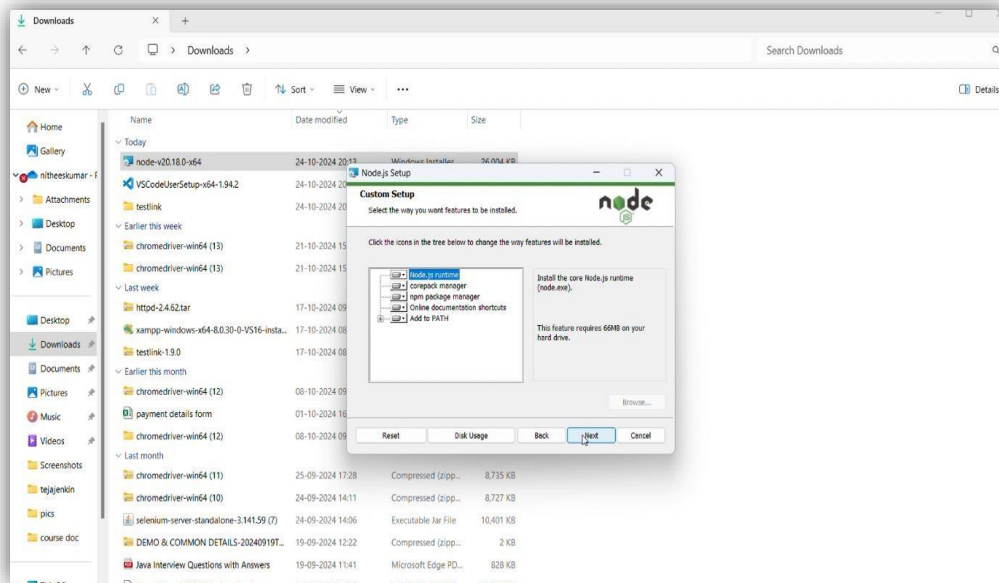
STEP 8: Click next option.



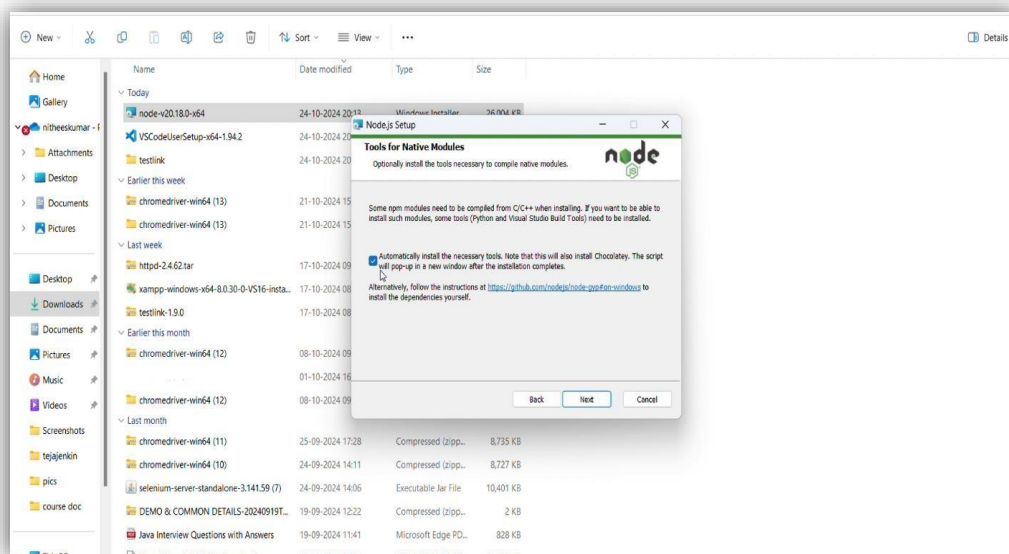
STEP 9: Click next to continue



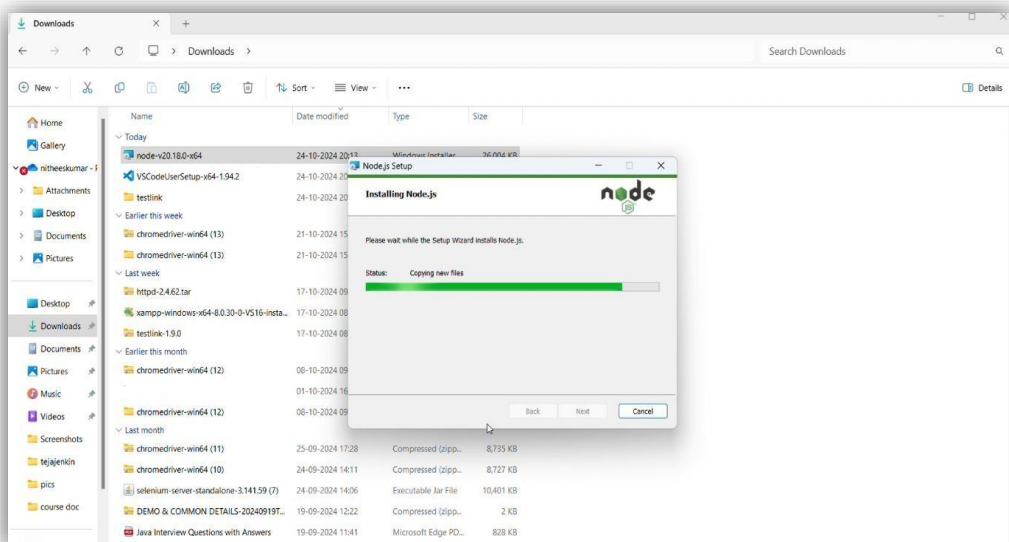
STEP 10: Click next to continue.



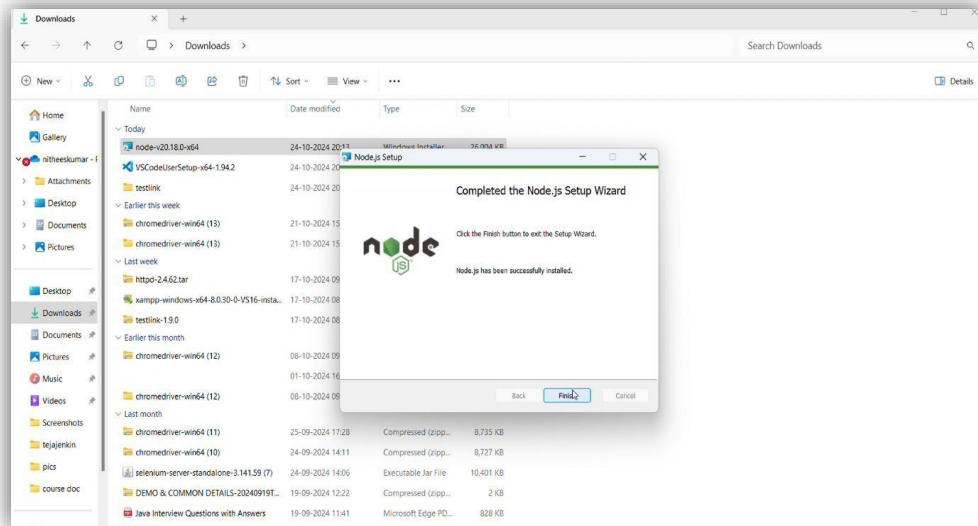
STEP 11: Click on the checkbox and click next.



STEP 12: Node js installing is in progress.

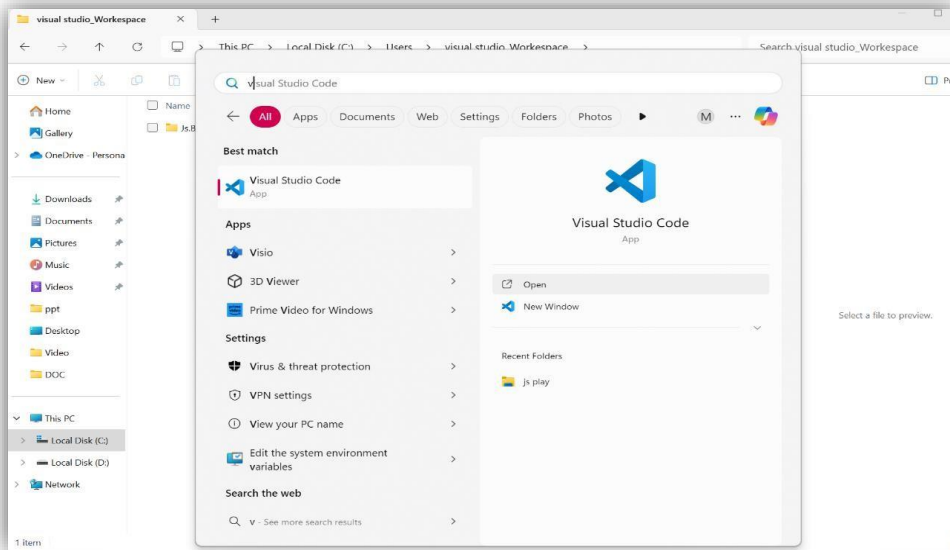


## STEP 13: Click Finish to complete

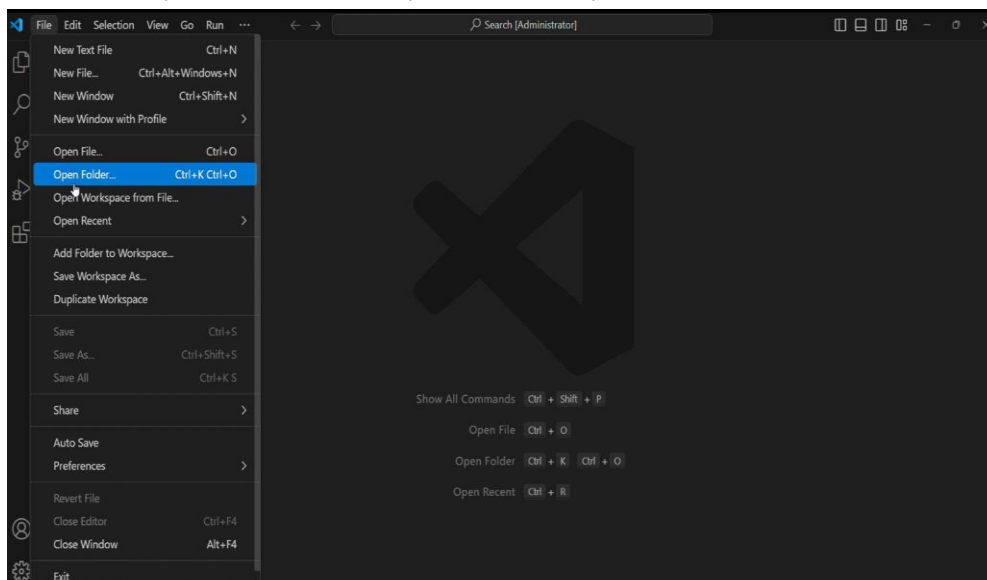


# VISUAL STUDIO CODE FILE CREATE & TYPESCRIPT INSTALLATION

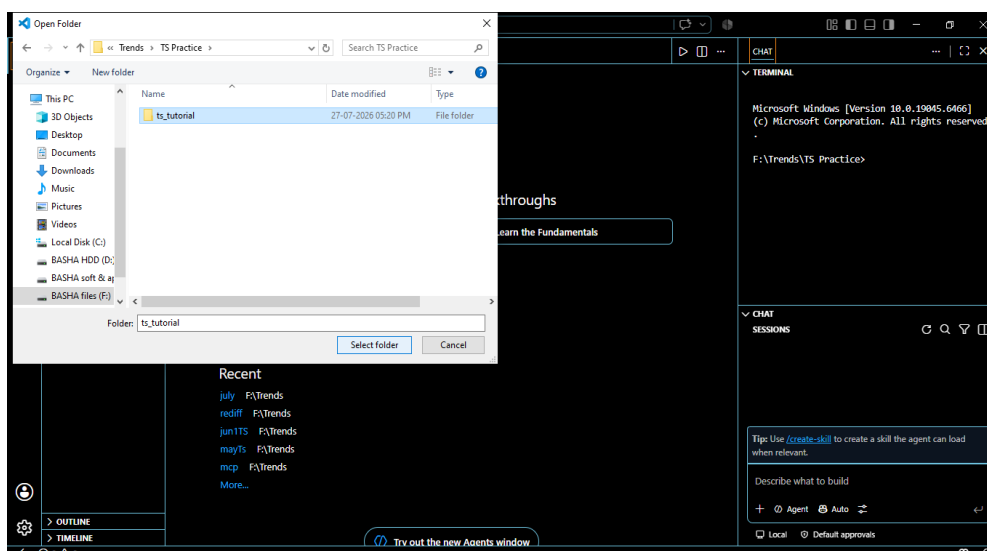
## STEP 1: Open the visual studio



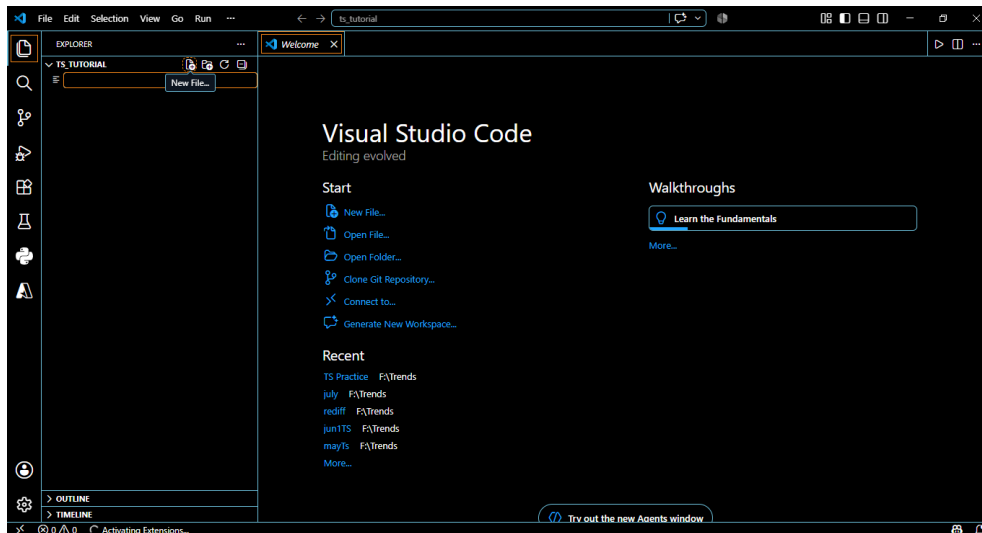
## STEP 2: Click the file option and click open folder option



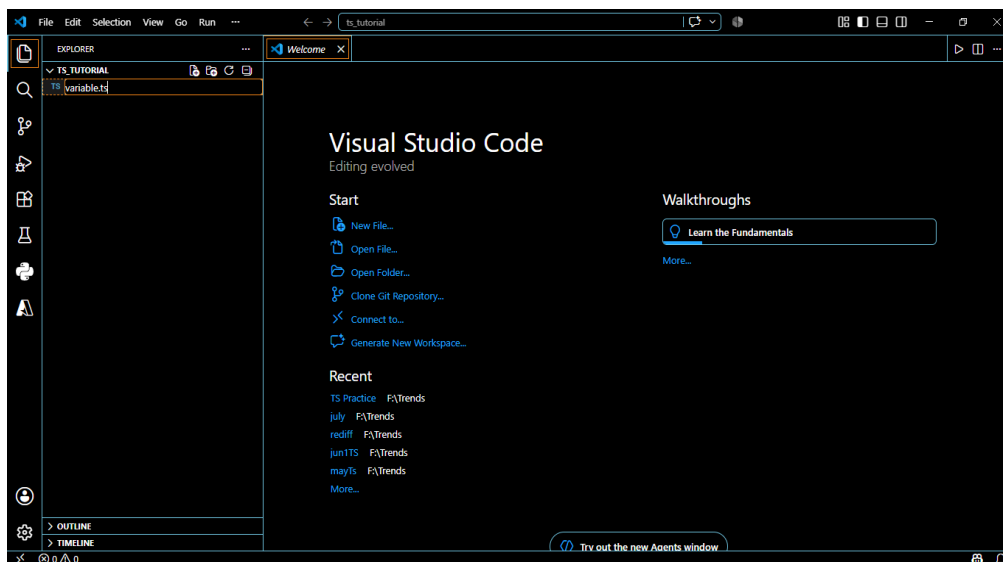
## STEP 3: Create or Select a folder



STEP 6: Click new file button besides the folder name.



STEP 7: Name file with (.ts) extension



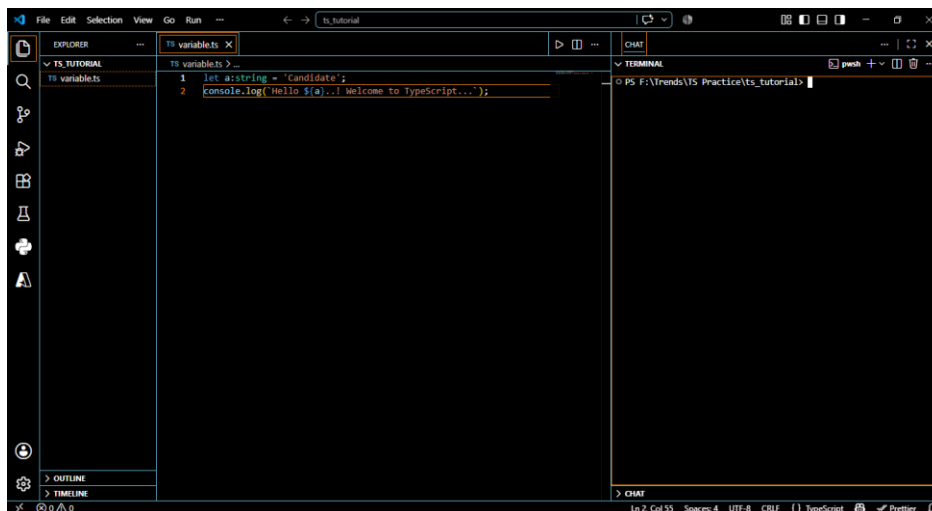
STEP 8: Start writing your code...



STEP 8: Click more in options, Select Terminal, Select New Terminal



STEP 9: In Terminal, use Command for Execution



# Thank You



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